

Custom NER Model

Identifying Key Entities in Recipe Data

Assignment Report

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Technique: Conditional Random Field (CRF)

Entity Classes: Quantity • Unit • Ingredient

This report is based on the executed outputs contained in the submitted Jupyter notebook. Charts, evaluation metrics, confusion matrices, class weights, and validation error analysis are reproduced from the notebook rather than estimated.

1. Executive Summary

The project builds a custom Named Entity Recognition (NER) system for recipe data. The task is to classify each token in a recipe into one of three labels: quantity, unit, or ingredient. A CRF model is trained using linguistic, lexical, quantity/unit, and contextual features.

The notebook starts with 285 recipes. Five records have mismatched token and label lengths and are removed, leaving 280 valid recipes. The cleaned data is split into 196 training recipes and 84 validation recipes. The model achieves 98.16% validation token accuracy with a 0.98 weighted F1-score. The main weakness is the unit class, which reaches 91.62% label accuracy and 0.93 F1.

Project Metric	Notebook Result
Original recipes	285
Invalid rows removed	5
Valid recipes	280
Training / validation	196 / 84
Training labelled tokens	7,114
Validation labelled tokens	2,876
Validation token accuracy	98.16%
Validation misclassified tokens	53
Weighted validation F1	0.98

2. Data Preparation and Quality

The source dataset contains two fields: input, representing recipe text, and pos, representing the corresponding token-level NER labels. The notebook validates that each input sequence has the same number of tokens as its label sequence.

Data-quality findings

Check	Result
Dataset shape	285 rows × 2 columns
Unique labels	ingredient, quantity, unit
Rows with unequal token/label lengths	5
Rows removed	17, 27, 79, 164, 207
Unequal rows after cleaning	0
Final valid recipes	280

The five invalid rows were removed because their input-token and label-token counts did not match. After cleaning, the notebook confirms that no unequal rows remain.

Train-validation split

A 70:30 split is applied, resulting in 196 training recipes and 84 validation recipes. The training set contains 7,114 labelled tokens and the validation set contains 2,876 labelled tokens.

3. Exploratory Data Analysis

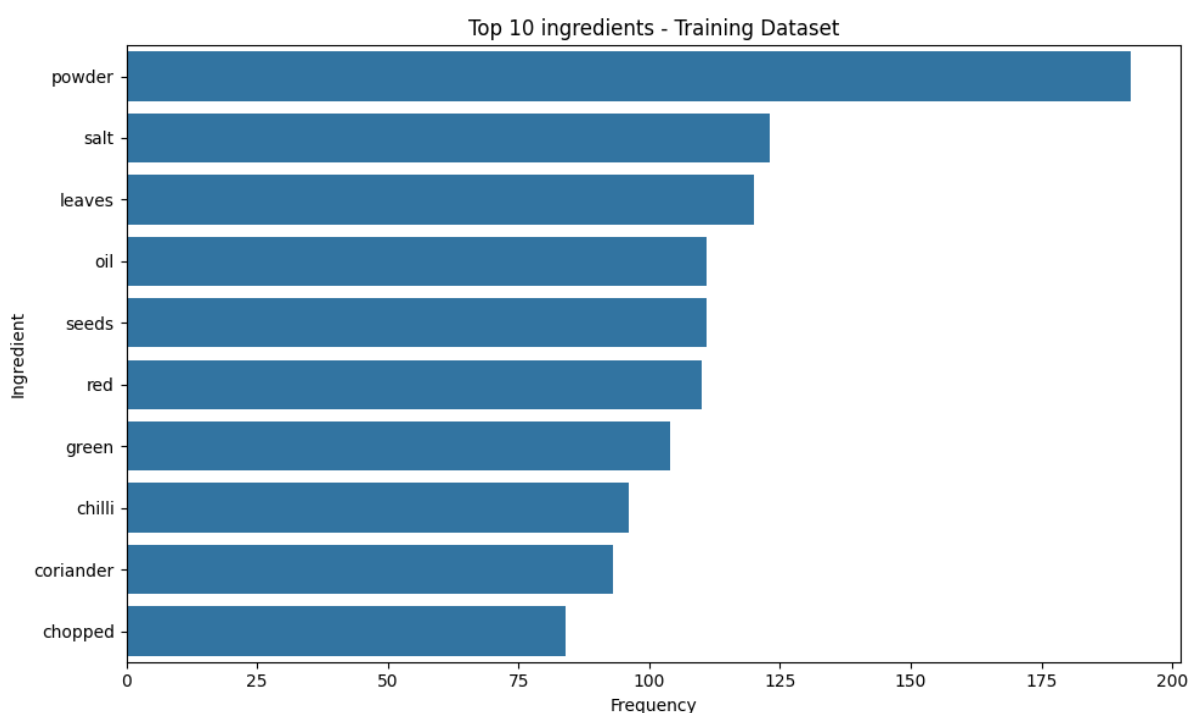
The exploratory analysis categorizes labelled tokens into ingredients, units, and quantities, then identifies the most frequent ingredients and measurement units in the training and validation sets.

Training-set token distribution

Entity type	Frequency
Ingredient	5,323
Quantity	980
Unit	811
Total	7,114

Ingredient tokens dominate the training data. This imbalance motivates the inverse-frequency class weighting used during CRF training.

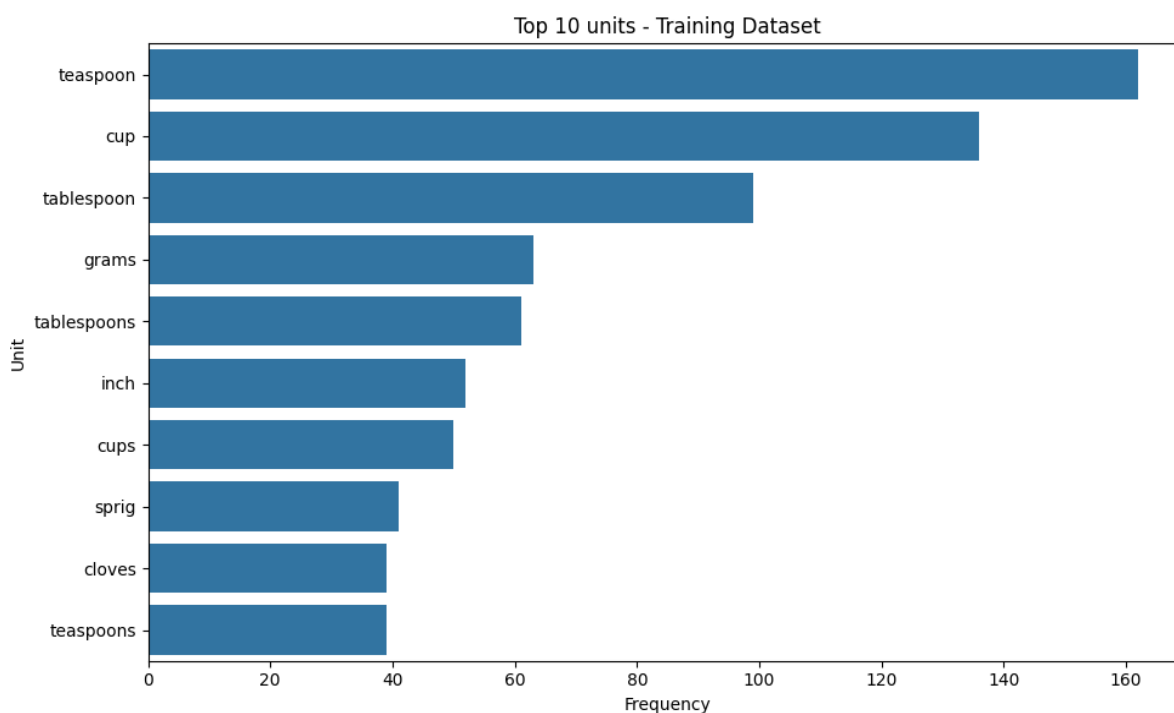
Top ingredients — training



The training data is led by powder, salt, leaves, oil, seeds, red, green, chilli, coriander, and chopped.

4. EDA — Measurement Units

Top units — training dataset



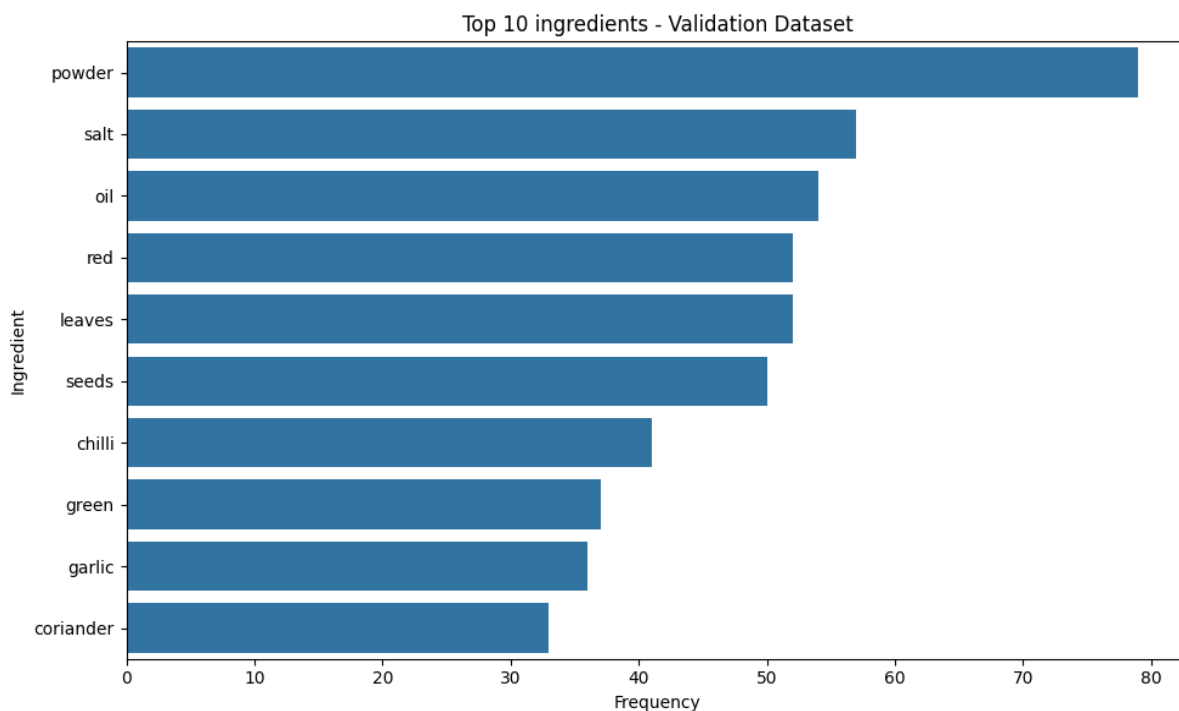
Teaspoon is the most frequent unit, followed by cup and tablespoon. Other common measurement expressions include grams, tablespoons, inch, cups, sprig, cloves, and teaspoons.

Training top-10 unit frequencies

Unit	Frequency
teaspoon	165
cup	136
tablespoon	100
grams	63
tablespoons	62
inch	52
cups	50
sprig	42
cloves	39
teaspoons	39

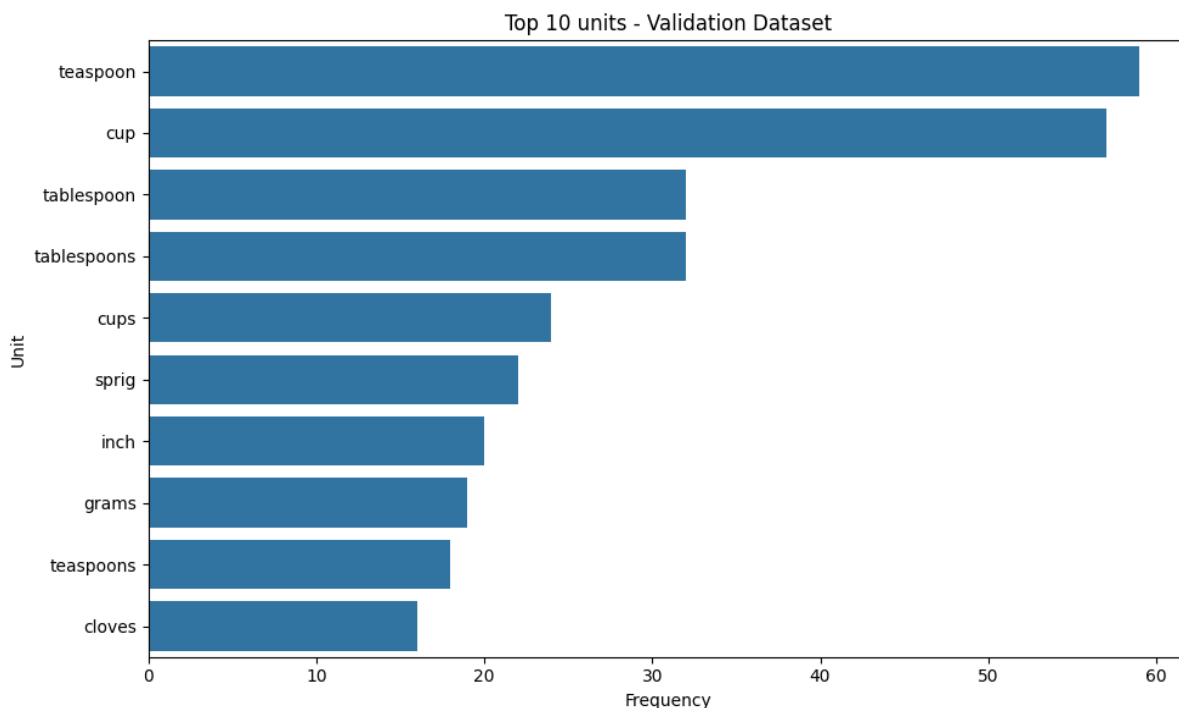
5. EDA — Validation Dataset

Top ingredients — validation



Powder, salt, oil, red, leaves, seeds, chilli, green, garlic, and coriander are the most frequent ingredient tokens in validation data. The overall pattern is consistent with the training set.

Top units — validation



Teaspoon and cup remain the most frequent units in validation data, suggesting that the train and validation splits have broadly similar measurement vocabulary.

6. CRF Feature Engineering

The notebook uses spaCy for linguistic processing and constructs token-level features for CRF. The feature design includes token lemma, part-of-speech information, token shape, quantity patterns, unit and quantity keyword indicators, and contextual features from neighboring tokens.

Class weighting

Label	Training support	Final class weight
ingredient	5,323	0.556860
quantity	980	2.419728
unit	811	2.923962

The inverse-frequency approach assigns larger weights to the minority classes. Unit receives the highest weight because it has the smallest training support.

CRF configuration

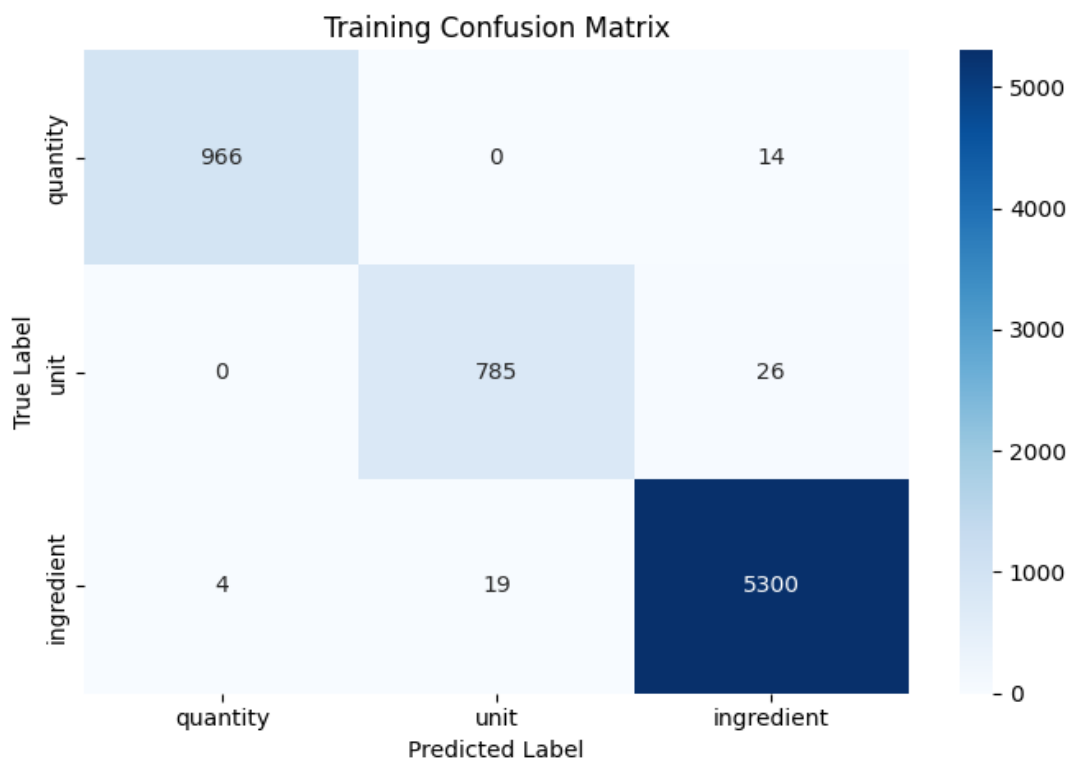
Hyperparameter	Value
Algorithm	lbfgs
c1	0.5
c2	1.0
Maximum iterations	100
All possible transitions	True

7. Training Performance

The trained CRF model performs strongly on the training data. The training classification report shows F1 scores of 0.99 for quantity, 0.97 for unit, and 0.99 for ingredient.

Label	Precision	Recall	F1	Support
quantity	1.00	0.99	0.99	980
unit	0.98	0.97	0.97	811
ingredient	0.99	1.00	0.99	5,323
Accuracy	—	—	0.99	7,114
Macro avg	0.99	0.98	0.99	7,114
Weighted avg	0.99	0.99	0.99	7,114

Training confusion matrix



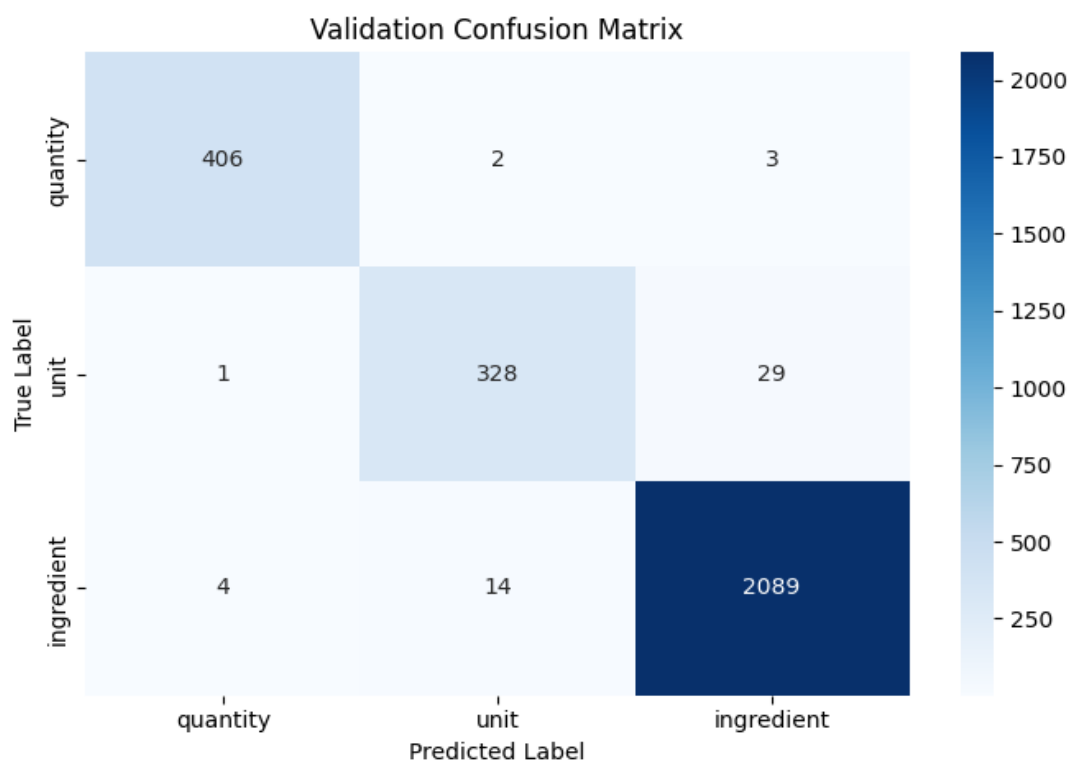
The training confusion matrix shows 966/980 correct quantity predictions, 785/811 correct unit predictions, and 5,300/5,323 correct ingredient predictions.

8. Validation Performance

Validation performance is the primary measure of generalization. The model achieves 98.16% overall token accuracy across 2,876 validation tokens, with 53 misclassified tokens.

Label	Precision	Recall	F1	Support
quantity	0.99	0.99	0.99	411
unit	0.95	0.92	0.93	358
ingredient	0.98	0.99	0.99	2,107
Accuracy	—	—	0.98	2,876
Macro avg	0.98	0.97	0.97	2,876
Weighted avg	0.98	0.98	0.98	2,876

Validation confusion matrix



The validation matrix shows 406 correct quantity predictions, 328 correct unit predictions, and 2,089 correct ingredient predictions. The largest error concentration is unit being predicted as ingredient.

9. Validation Error Analysis

Label	Support	Correct	Label accuracy	Errors
ingredient	2,107	2,089	99.15%	18
quantity	411	406	98.78%	5
unit	358	328	91.62%	30

The notebook identifies 53 misclassified validation tokens. Unit accounts for 30 of these errors, making it the main opportunity for improvement. Many mistakes involve context-sensitive terms such as cloves, pieces, inch, sprig, slices, spoon, and descriptive/cooking words.

Representative examples

Token	True	Predicted	Context
few	ingredient	quantity	Tamarind Leaves few
cloves	ingredient	unit	chillies 3 cloves garlic big
Spoon	unit	ingredient	garlic big Spoon oil teaspoon
pieces	ingredient	unit	into small pieces 1/4 peas
pinch	quantity	unit	Urad Dal pinch Asafoetida hing
to	quantity	ingredient	Cane sugar to tablespoons Extra

These errors demonstrate that difficult cases often depend on how a word is used in surrounding recipe context.

10. Main Findings and Insights

1. Strong overall performance. The CRF reaches 98.16% validation token accuracy and 0.98 weighted F1, showing that the engineered features are effective for this dataset.
2. Ingredient recognition is highly reliable. Ingredient label accuracy is 99.15% on validation data, with 2,089 of 2,107 tokens correctly identified.
3. Quantity recognition is also strong. Quantity achieves 98.78% label accuracy and 0.99 F1.
4. Unit recognition is the main challenge. Unit accuracy is 91.62% and F1 is 0.93. The majority of remaining validation errors are associated with this class.
5. Class weighting addresses imbalance. Ingredient is the dominant class, while unit and quantity receive larger weights.
6. Context matters. Many errors occur for words that can look like ordinary ingredients or descriptive words unless neighboring tokens are considered.

11. Conclusion

The completed CRF-based NER pipeline successfully converts recipe text into structured quantity, unit, and ingredient entities. The model demonstrates strong generalization on the validation set and provides a practical foundation for structured recipe databases, recipe search, meal planning, and grocery-assistance applications.

The clearest next improvement is to strengthen unit recognition through additional contextual features, more comprehensive unit normalization, and additional examples of ambiguous measurement terms. Overall, the notebook demonstrates a successful end-to-end custom NER solution using CRF.

12. Assumptions and Limitations

- Evaluation is token-level rather than recipe-level exact-match evaluation.
- The report uses the executed notebook outputs as the source of truth.
- The dataset contains noisy/descriptive recipe text, which creates genuine ambiguity.
- Results are specific to the supplied dataset, split, feature engineering, and CRF configuration.